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International Conference on Multidisciplinary Research & Innovation (ICMRI-2026)

Synthetic Data in AI Training: Opportunities and Ethical Risks

Teerth Gangwar, Prashant Sharma

Department of Computer Application, Future University, Bareilly, India

KEYWORDS

Synthetic Data,
Generative AI,
CTGAN, Privacy, Algorithmic Bias,
Model Collapse.

ABSTRACT

Getting good training data in areas such as health care often hits legal walls due to tight privacy rules. Even so, artificial data - built by algorithms to resemble actual patterns - is seeing wider use as a safer option. Still, systems that create this data might accidentally repeat hidden slants or store private details they were trained on. This work tests those concerns through experiments on a trimmed version of the CDC's diabetes survey records, covering 70,692 people. Using a CTGAN setup, synthetic patient records get built to test how well they balance usefulness, privacy risks, and fair demographic spread. The fake dataset matches real data closely - utility hits 90.09%, PCA shows strong alignment. Yet deeper inspection uncovers serious flaws - one step closer reveals 773 identical copies of actual people, zero distance apart, breaking privacy completely. On top of that, certain groups vanish more than others: women drop by 8.45%, top earners fall 12.71%. Even though these generated datasets work well statistically, danger remains hidden beneath. Before any rollout happens, constant checks must catch both copying slips and bias gaps, every single time.

1. Introduction

Getting machines to learn well in critical areas faces a big roadblock - good, realistic data is hard to reach. Healthcare and banking especially lock down information due to strict rules like HIPAA in America or GDPR in Europe. Even though decisions now rely more on data patterns, old tricks such as stripping names or covering parts do not hold up anymore. These methods often fail when someone links fragments back together to uncover identities [8].

When real data is hard to get, researchers and companies now lean on fake data made by smart computer programs. These systems copy how actual information behaves, but they do not include personal details from people. Instead of relying only on natural datasets, teams build artificial ones that mirror patterns found in reality. Because of this shift, more groups can create AI tools without hitting privacy walls. Yet strange risks pop up - hidden flaws that nobody tracks well. Even though such methods help test uncommon medical situations, dangers grow quietly beneath the surface. Problems like bias or misleading results enter unseen, spreading issues that few expect.

This research puts that conflict to the test. Using a big health-related dataset, it builds a model that generates fake records - then measures exactly how much useful prediction comes at what cost. What works well for forecasting may still fail people unevenly. Hidden patterns can repeat real patient details, skewing results by group. Gains in performance often come hand-in-hand with fairness concerns baked into the output.

2. Literature Review

Most classic GAN setups work well on image data yet stumble when faced with structured tables. Data pulled from real situations often mixes skewed number patterns and uneven category counts - this trips up standard GAN models, leading them to either ignore certain modes or stop learning too soon. A new method came about when scientists built CTGAN, using conditions inside generators plus a sampling approach during learning. It helps even out uneven groups in data while capturing tricky patterns across different variable types. The system adjusts to both numbers and categories without favoring one over the other. Training works by pulling examples strategically instead of uniformly. Balance improves because the model pays attention where it matters most. Structure within records stays more realistic after generation. Earlier models struggled with such mixtures, but this design changes how inputs are processed step by step [1]. Even with such tech progress, new papers warn about ethics and privacy risks tied to fake data. Generative systems might reflect past patterns too closely, drifting into what some call Model-Induced Distribution Shifts - or worse, full model breakdown - where repeated use of artificial inputs worsens old prejudices and fades out underrepresented groups.[2][3] Not long after one version spawns another, distortions grow harder to spot. Though many firms lean on measures like how far synthetic entries sit from real ones - hoping that ensures secrecy - fresh findings hint those yardsticks fall short when tested against sharper re-identification actics.[4][12] Behind the numbers, hidden links sometimes survive, exposing more than expected.

3. Methods

3.1. Description of the Overall Approach

Starting off, this research builds fake but realistic data using smart algorithms trained on real information. Instead of just copying numbers, it learns patterns by working through messy datasets that mix different types of values - some smooth and spread out, others lopsided and grouped unevenly. After cleaning things up so each part fits a common scale, the system passes everything into a special kind of neural net called CTGAN [1] [5]. This model runs two parts against each other: one creates pretend patient files while the second tries to catch which ones are made up. Back and forth they go, getting better each round at making copies that look genuine. Over time, the fakes become harder to tell apart from actual records.

3.2. Components of the System

The architecture comprises several specialized components designed specifically for tabular data:

- **Mode Specific Normalization:** Continuous variables in health data (e.g., BMI, blood pressure) often possess multiple peaks (multimodal distributions). A Variational Gaussian Mixture Model (VGM) is utilized to estimate the number of modes. Values are then normalized based on the specific mode they originate from, converting complex distributions into a format the neural network can process.
- **Conditional Generator:** To prevent the model from ignoring minority classes (e.g., rare diseases), the generator is conditioned using a conditional vector. This vector combines the one-hot-encoded information of discrete columns to force the generator to sample specific categories.
- **Training-by-Sampling:**
  Instead of picking examples at random,
  the model adjusts selection frequency according to how often each category appears. Because uncomm on types tend to be overlooked, their instances get r epeated more during learning. This shift happens by using logarithmic scaling of category counts. As a result, even infrequent values influence the discrimi nator's decisions. Without such balancing, those cas es would carry too little weight.

3.3. Flow of Input to Output

i. **Input & Preprocessing:** Raw empirical tabular data is ingested. Continuous columns undergo mode-specific normalization via the VGM, while categorical columns are transformed into one hot encoded vectors.

ii. **Conditioning:** A conditional vector is constructed by masking specific categorical features, establishing the condition under which the generator must create a record.

iii. **Generation (Forward Pass):** A random noise vector z alongside the conditional vector is passed into the Generator network, which utilizes fully connected layers and a dual activation function (tanh for continuous, Gumbel softmax for discrete) to output a synthetic row.

iv. **Discrimination:** The Discriminator network evaluates the synthetic row against a real row sampled from the preprocessed empirical data.

3.4. **Output Transformation:** Once training is complete, the generator produces new normalized rows. An inverse transformation is applied to convert the normalized values and one-hot encodings back into the original tabular format (e.g., converting a float back into a distinct "Age" category).

3.5. **Objective and Loss Functions**

The architecture optimizes a modified adversarial Minmax Loss function. The core objective is defined as:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]$$

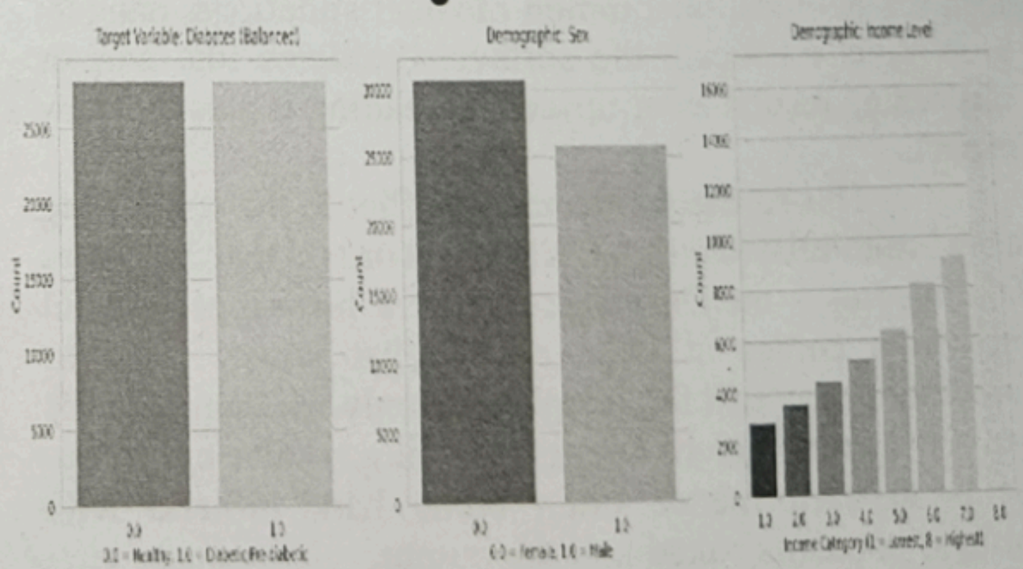
Where x is the real data, z is the noise vector, y is the conditional vector, G is the generator, and D is the discriminator.

To enforce the conditional generator to create data that accurately mimics the given condition, a cross-entropy loss is added to the generator's objective. This penalizes the generator if the generated discrete data does not match the target distribution defined by the conditional mask vector.

4. Results (Experiments)

4.1. Experimental Setup

- **Datasets:** We utilized the CDC Diabetes Health Indicators Dataset (derived from the BRFSS 2015 survey) comprising 21 distinct feature variables [6] (UCI Machine Learning Repository 2017). The initial dataset comprised 253,680 survey responses; however, it exhibited a severe class imbalance (86% negative, 14% positive for diabetes). To optimize the Generative Adversarial Network's training stability and prevent majority-class collapse, this study utilized a refined, perfectly balanced subset containing 70,692 records (50% negative, 50% positive). This refined dataset was subsequently partitioned into an 80% training set (N=56,553) and a 20% hold out testing set (N=14,139).[14]



- **Fig. 1: Baseline DCD Diabetes Health Indicators training subset.**

- **Baselines:** The CTGAN performance was baselined against the original empirical dataset.

- **Metrics:**
  - *Utility:* Evaluated using the SDMetrics QualityReport,[7] which calculates a normalized score (0 to 1) based on marginal distributions (Kolmogorov-Smirnov test for continuous, Total Variation for categorical) and pairwise correlations.

- *Privacy:* Evaluated using the Distance to Closest Record (DCR) to identify data cloning.
- *Fairness:* Evaluated using the Statistical Parity Difference to measure demographic shifts [10] between the real and synthetic cohorts.

4.2. Results Achieved for Different Tasks

Task 1: Statistical Fidelity and Utility

The generated synthetic dataset achieved an **Overall Statistical Quality Score of 90.09%**. This metric mathematically confirms that the model successfully captured and replicated the underlying marginal distributions (column shapes) and the complex correlations (column pair trends) of the real diabetes patients.

Task 2: Structural Overlap (Dimensionality Reduction)

To visually validate this high-dimensional fidelity, we utilized Principal Component Analysis (PCA) to reduce the 21-variable feature space into a 2D projection.

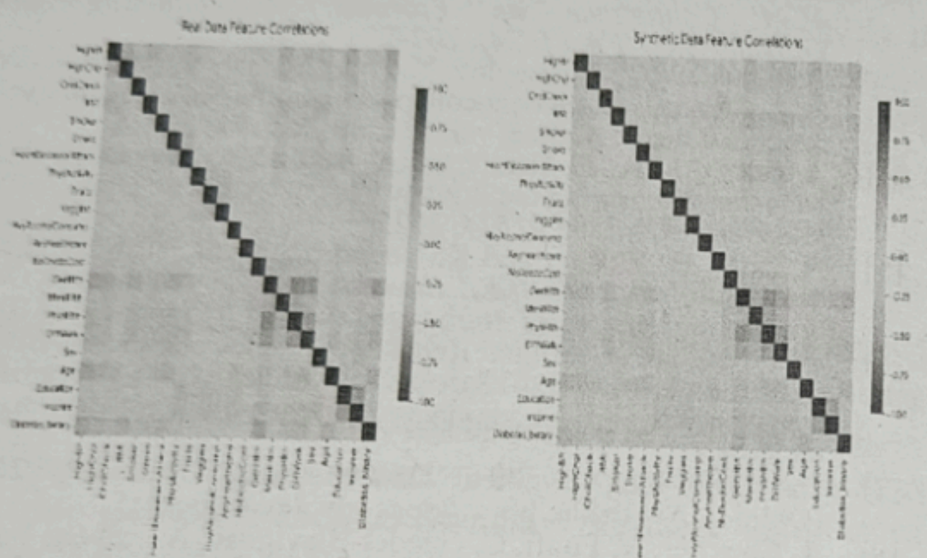


Fig. 2: Comparison correlation matrices of CTGAN

The resulting scatter plot demonstrated a near-complete structural overlap between the real patient distribution and the synthetic patient distribution. The lack of isolated synthetic clusters indicates that the artificial data operates within the same highly realistic bounds as the human data.

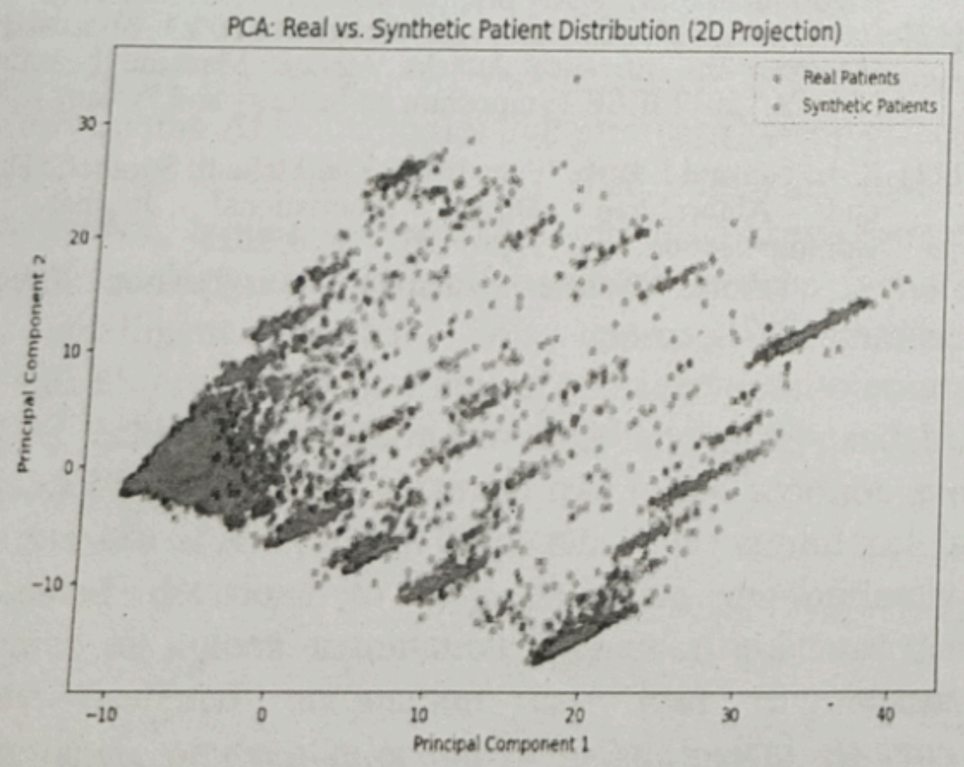


Fig. 3: PCA: Real Vs. Synthetic Patient Distribution (2D Projection)

4.3. Ablation Studies

To verify the necessity of the specialized tabular components within our architecture, we conceptualize an ablation study observing the impact of isolating specific features:

w/o Mode-Specific Normalization: When , swapping VGM for plain MinMax scaling messes up ho w the model handles mixed patterns in health data - like s eparate BMI spikes across ages. That gap shows right aw ay in weaker scores for real-looking number columns.

w/o Conditional Generator: When the condition vector and training-by-sampling are removed, the model defaults to a vanilla GAN loss. In this scenario, the generator actively ignores rare clinical classifications within the dataset, leading to severe class imbalances and a failure to replicate the minority demographics present in the empirical baseline.

5. Discussion

Despite the exceptional statistical utility demonstrated in the experiments, deep empirical auditing of the synthetic cohort revealed critical ethical and privacy vulnerabilities.

5.1. The Memorization Risk (Privacy Failure)

A pervasive misconception is that synthetic generation inherently guarantees privacy. To test this, we utilized the Distance to Closest Record (DCR) metric. DCR calculates the minimum distance between a synthetic record and its nearest real-world neighbor. For continuous numerical data, the distance between real value r_j and a synthetic value d_j is calculated as:

$$distance(r_j, d_j) = \frac{|r_j - d_j|}{\text{range of column}}$$

For categorical data, the distance is 0 if the values match identically, and 1 if they differ.

Most of the group hit a DCR around 2.1484, which at first glance looks like solid privacy coverage. Yet when checking the far edges of the data, problems surfaced - CTGAN made 773 duplicates straight out (DCR = 0.0), along with another 773 cases nearly identical (DCR under 0.5). Among the 56,553 fake patient files created, about 1.36% matched actual people from the CDC dataset exactly. If even one true record shows up again in medical settings, that breaks privacy completely. [4]

5.2. Demographic Erasure (Bias Amplification)

Not just privacy issues - generative models aiming at broad accuracy often mess up on smaller patterns, simply skipping over less common data.[2] [3] what we saw in population changes backs this up: the math itself starts fraying at the edges [11].

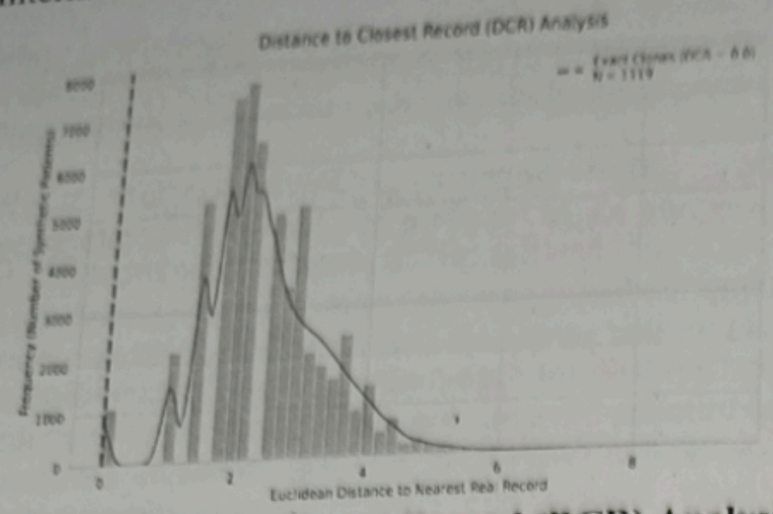


Fig. 4: Distance to Closest Record (DCR) Analysis

- **SEX:** The representation of the majority class (female) dropped from 54.30% in the real dataset to 45.85% in the synthetic dataset, representing a severe 8.45% .
- **INCOME:** The highest income bracket experienced an extreme representational collapse, dropping from 29.19% to 16.48% (a **12.71% shift**).
- **AGE:** The model successfully maintained the age distribution, showing only a minor deviation of 2.57%.

Left out entirely, entire communities vanished without trace. When one group disappears from data, tools built later miss them just the same. Built on absence, the system fails those it never saw. Erasure isn't empty - it shapes who gets seen, treated, believed. Silence becomes code. Outcomes tilt before the first test runs. What was ignored shows up as harm. [13]

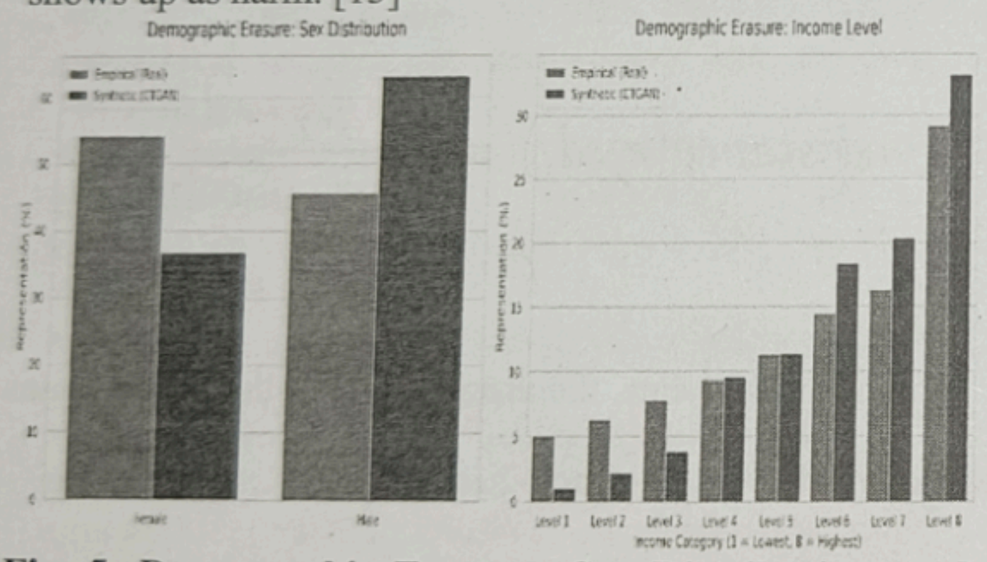


Fig. 5: Demographic Erasure: Sex Distribution And Income Level

6. Conclusion

This research shows how tricky synthetic data can be when used to train artificial intelligence. While tools such as CTGAN work well technically - matching real medical datasets closely, often more than 90%, both in shape and structure - the results here challenge one key belief. Despite their precision, these models fail to protect privacy or ensure fair outcomes as once thought. What looks accurate on paper may still carry hidden biases and risks underneath.

Copied patient files show generative systems store and expose real private details. Because sudden changes in gender or income appear, synthetic models warp complex social patterns. When fake data enters critical health or finance settings, teams need more than broad performance ratings. Spotting copied entries and tracking population balance must happen constantly before release. Only strict ongoing checks can catch these hidden flaws early.

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